OTA-C 50/60 Hz Notch Filter in Sky130

Bulk-driven OTA style: design, sizing, verification, power, timing, OTA layout and post-layout analysis

This is the second of three OTA styles. It reuses the filter, capacitors, buffer and testbench from the current-cancellation report, so those sections are summarised here and the new material is the OTA itself. Every number comes from a simulation in this project or a calculation shown here.

1. Result in one page

Quantity	Result	Target	Status
Notch frequency (tuned bias)	49.97 Hz at Ib = 0.3382 nA; 60.01 Hz at Ib = 0.4133 nA	50 / 60 Hz within 1 Hz	Met
Notch depth (vs 10 Hz gain)	38.2 dB at 50 Hz; 38.1 dB at 60 Hz	≥ 20 dB	Met
Low-frequency gain	-4.00 dB	about −6 dB	Higher than ideal
Gain above the notch	-6.62 dB at 100 Hz, -7.17 dB at 150 Hz; upper −3 dB edge 79.6 Hz	Not specified	Better than style 1
Stability (2 mV step)	Settles in 40.8 ms, 15.6% overshoot, no growth	No oscillation	Met
Mains rejection in time	50 Hz is 38.6 dB below 10 Hz	≥ 20 dB	Met
THD at 5 mV, 10 Hz	0.81% at the output; 0.07% at the filter output	≤ 2%	Met
THD at 10 / 20 mV	1.61% / 3.22% at the output (buffer-limited)	≤ 2%	**20 mV fails**
Mismatch Monte Carlo (100 runs)	100% have depth ≥ 20 dB; f_0 50.25 ± 3.15 Hz; 92% within ±5 Hz	Reported	**No failures**
Process corners and temperature	f_0 44.9–55.1 Hz at fixed bias; retrimmed to 49.7–50.4 Hz	Reported	Tunable
Power	1.574 µW total: filter 4.10 nW, buffer 1.565 µW	Reported	Buffer dominates
OTA layout	90.2 × 27.4 µm; DRC 0 errors; LVS: circuits match uniquely	Clean	Met
Post-layout	49.96 Hz, 34.3 dB at Ib = 0.3418 nA; Monte Carlo 100/100 usable	Close to schematic	Met, 4 dB shallower

Honest summary: compared with current cancellation, this style survives mismatch (no Monte Carlo failures, four times the linear range) but spreads the notch frequency more, has twice the input noise, and its THD above about 12 mV is set by the shared buffer. Layout costs about 4 dB of notch depth because each input drives its own n-well.

2. What is shared with the first style

The filter is the 3rd-order elliptic OTA-C notch (five OTAs, $C_2 = C_{L2} = 4.084$ pF and $C_1 = C_3 = 0.524$ pF MiM capacitors). Its transfer function, the stability argument against the 6th-order filter, and the buffer design are derived in full in the current-cancellation report. The key relations are:

$$\omega_0 = g_m / \sqrt{(C_2 C_{L2})} \quad H(0) = g_{m1} / (g_{m4} + g_{m1} g_{m5} / g_{m3}) = 1/2 \text{ when all } g_m \text{ are equal}$$

So the target is again $g_m = 2\pi \times 50\ \text{Hz} \times 4\ \text{pF} \approx 1.26$ nA/V, tuned by the bias current.

3. How a bulk-driven transistor works

A MOSFET has a fourth terminal, the body (bulk). Changing the body-to-source voltage shifts the threshold voltage, so it also changes the drain current. The body is a weaker control than the gate. In weak inversion:

$$g_{mb} = \eta \cdot g_m, \text{ with } \eta = (n - 1)/n \approx 0.28 \text{ for the Sky130 PFET measured earlier } (n = 1.81)$$

Using the body as the input therefore divides the transconductance by about 3 to 4 at the same current. That is exactly what an ultra-low-g_m OTA wants. It also widens the input range by the same factor, because the input must move $1/\eta$ times further to produce the same current change.

4. Adapting the paper's circuit to Sky130

The 32 nm paper drives the bodies of an NMOS pair at ±0.15 V. Two changes were needed:

• **PMOS inputs.** An NMOS body is the shared p-substrate unless the device sits in an isolated p-well inside a deep n-well. A PMOS body is simply its own n-well, which Sky130 provides directly. The circuit was flipped to PMOS inputs with NMOS diode loads and mirrors; it works identically with polarities exchanged.

• **Replica gate bias.** The paper ties the input gates to ground. Here the input source would then sit about 0.25 V above the body, forward-biasing the source-body junction. A replica PMOS diode, the same size as one input device, carries Ib/2 with its source at 0 V and sets the gate voltage v_g. Each input device then has the same current density as the replica, so its source also sits near 0 V, level with its body. The replica tracks process and temperature automatically.

Signal path: the inputs drive the bodies of M1 (inn) and M2 (inp); M3/M4 are diode loads; M4 mirrors to M5, M3 to M6, M6 feeds the PMOS mirror M7/M8, and

$$I_{out} = I(M8) - I(M5) = I(M1) - I(M2)$$

Raising inp raises the threshold of M2 and cuts its current, so I_{out} rises: inp is the non-inverting input.

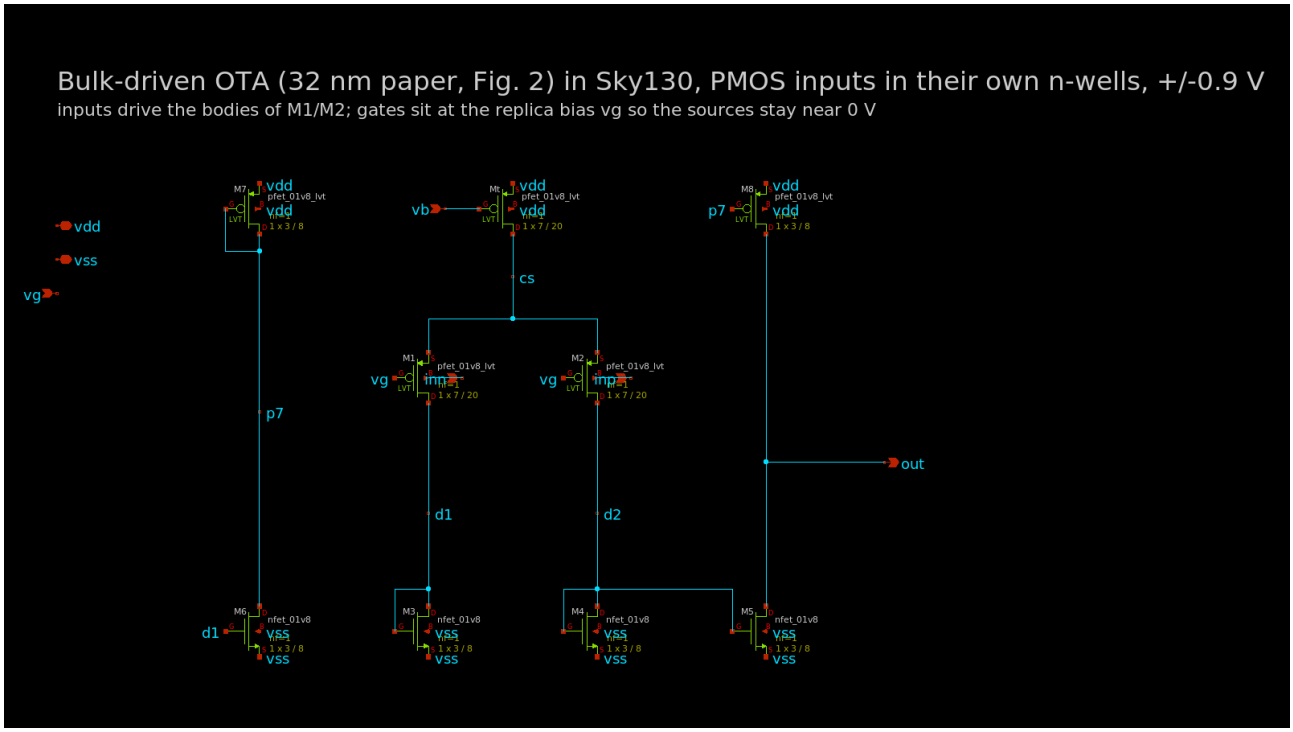


Figure 1. Bulk-driven OTA in Xschem: input transistors M1/M2 with their bodies as inputs, NMOS loads and mirrors, PMOS output mirror.

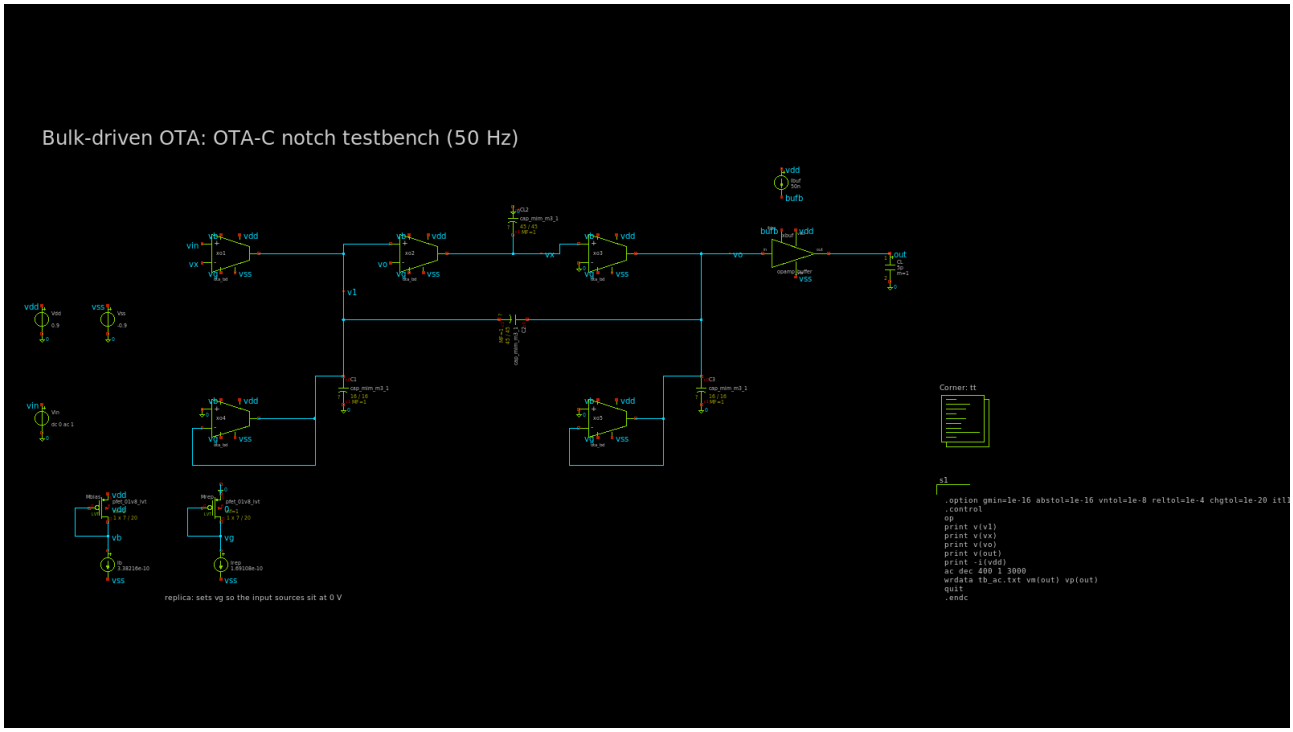


Figure 2. Notch testbench with five bulk-driven OTAs, MiM capacitors, the tail-bias diode, the vg replica and the buffer.

5. Sizing

The OTA transconductance is the body transconductance of one input device carrying half the tail current:

$$g_{m,OTA} = (I_b/2) \cdot (g_m/I_D) \cdot \eta$$

With g_m/I_D = 24.0 /V for the input devices and η = 0.285 (the standard-PFET value), the hand estimate at I_b = 0.36 nA is

$$g_{m,OTA} = 0.18\ \text{nA} \times 24.0\ \text{/V} \times 0.285 = 1.23\ \text{nA/V}$$, versus 1.468 nA/V from simulation

The simulation implies $\eta \approx 0.34$ for the low-Vt PFET actually used, so the hand number is 16% low, and the final bias came out lower than first estimated. Device sizes reuse the lessons from the first style: large input devices (W/L = 7/20 µm) for matching, small mirrors (3/8 µm) so their gate capacitance does not add phase at 50 Hz.

Device	Role	Type	W/L (µm)
Mt	Tail current source	pfet_01v8_lvt	7/20
M1, M2	Bulk-driven input pair	pfet_01v8_lvt	7/20
M3, M4	Diode loads	nfet_01v8	3/8
M5, M6	Mirrors	nfet_01v8	3/8
M7, M8	Output mirror	pfet_01v8_lvt	3/8
Mbias	Tail bias diode (shared)	pfet_01v8_lvt	7/20
Mrep	vg replica (shared)	pfet_01v8_lvt	7/20

OTA performance in two runs: the first sizing run at I_b = 0.36 nA, and the validation chain at 0.31 nA (Figure 3), against the current-cancellation OTA:

Quantity	Bulk-driven, 0.36 nA	Bulk-driven, 0.31 nA	Current-cancellation, 0.209 nA
g_m	1.468 nA/V	1.284 nA/V	1.529 nA/V

Quantity	Bulk-driven, 0.36 nA	Bulk-driven, 0.31 nA	Current-cancellation, 0.209 nA
Linear range (±10% g_m)	−99 to +97 mV	−96 to +95 mV	−23 to +25 mV
Output resistance	27.8 GΩ	32.2 GΩ	49.95 GΩ
Open-loop gain	32.2 dB	32.3 dB	37.7 dB
CMRR	72.4 dB	72.3 dB	55.9 dB
Systematic input offset	−18.1 mV	−18.1 mV	7.6 mV
Input noise, 0.05–300 Hz	312.6 µV rms	331.8 µV rms	149.9 µV rms
Power, OTA + bias diode	2.11 nW	1.82 nW	0.76 nW

Offset and noise are both input-referred through the weak body control, so they grow by roughly 1/η. The offset is still only 18% of the linear range, against 30% for the current-cancellation OTA, which is why this style survives mismatch better.

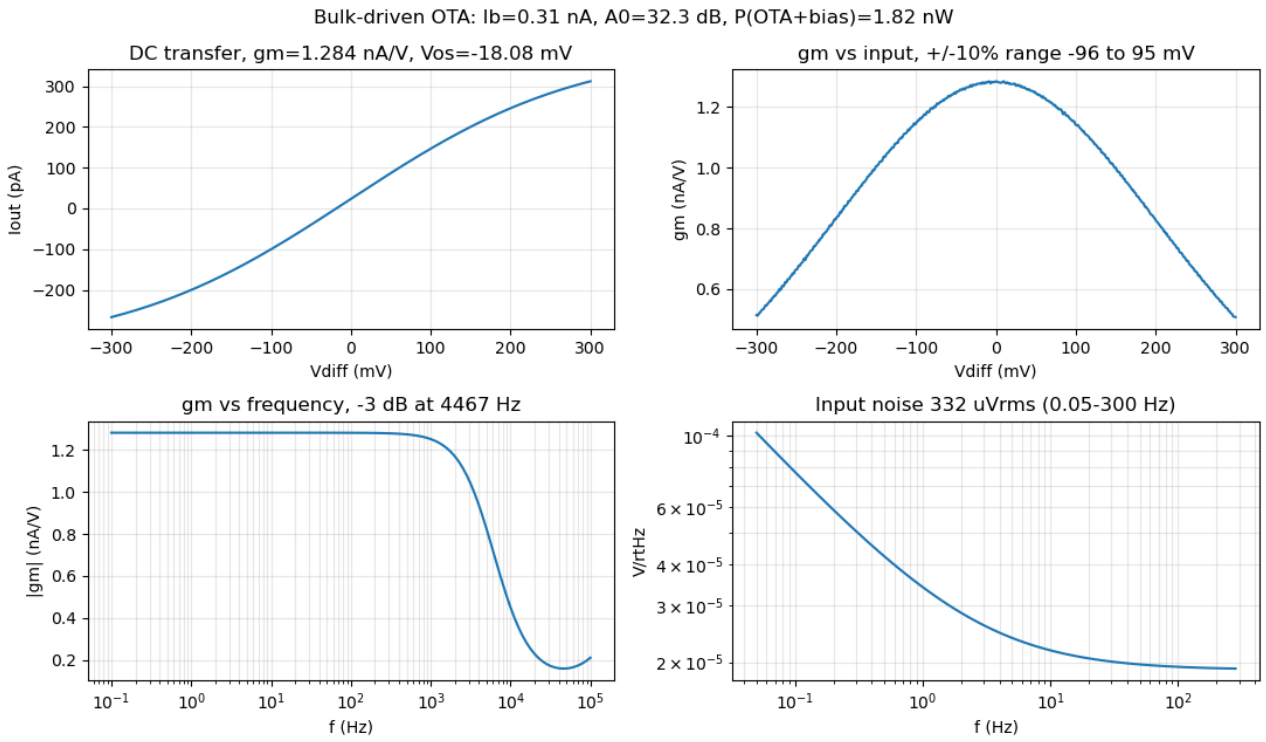


Figure 3. OTA validation at I_b = 0.31 nA: DC transfer, g_m vs input (note the wide flat region), g_m vs frequency, input noise.

6. Filter results

Tuning needed I_b = 0.3382 nA for 49.97 Hz and 0.4133 nA for 60.01 Hz, a ratio of 1.222 (ideal 1.2). The Xschem netlist (39 devices, 0 errors) reproduces the result exactly.

Frequency (Hz)	1	10	25	40	75	100	150	300
Gain (dB)	-3.96	-4.00	-4.32	-7.47	-7.26	-6.62	-7.17	-9.75

Two differences from the first style:

• **The passband sits at −4.0 dB instead of −7.1 dB.** The likely cause is unequal g_m across the five OTAs at their DC operating points. The OTA's systematic offset moves the termination OTA xo5 about 44 mV off balance, where its g_m is lower, and by $H(0) = g_{m1}/(g_{m4} + g_{m1}g_{m5}/g_{m3})$ a lower g_{m5} raises the gain. I have not proven this.

• **The response recovers above the notch** (upper −3 dB edge 79.6 Hz), because the smaller mirrors and lower output resistance add less parasitic loading at the high-frequency poles.

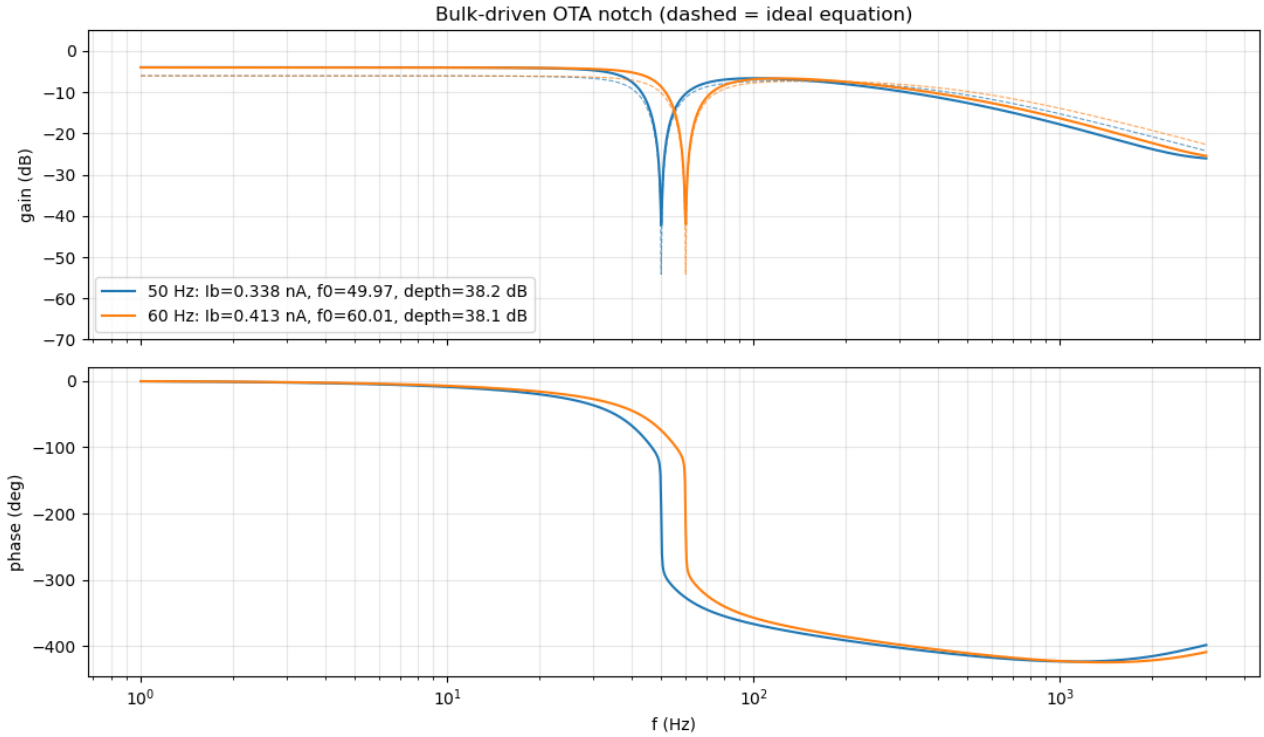


Figure 4. Notch response at the buffer output, tuned to 50 Hz and 60 Hz.

Transient. A 2 mV step settles to 1.277 mV (consistent with the −4 dB gain) in 40.8 ms with 15.6% overshoot. After settling only 7–10 µV of flat solver ripple remains, which does not grow. In the 2 mV + 2 mV mix of 10 Hz and 50 Hz, the 50 Hz tone comes out 38.6 dB below the 10 Hz tone.

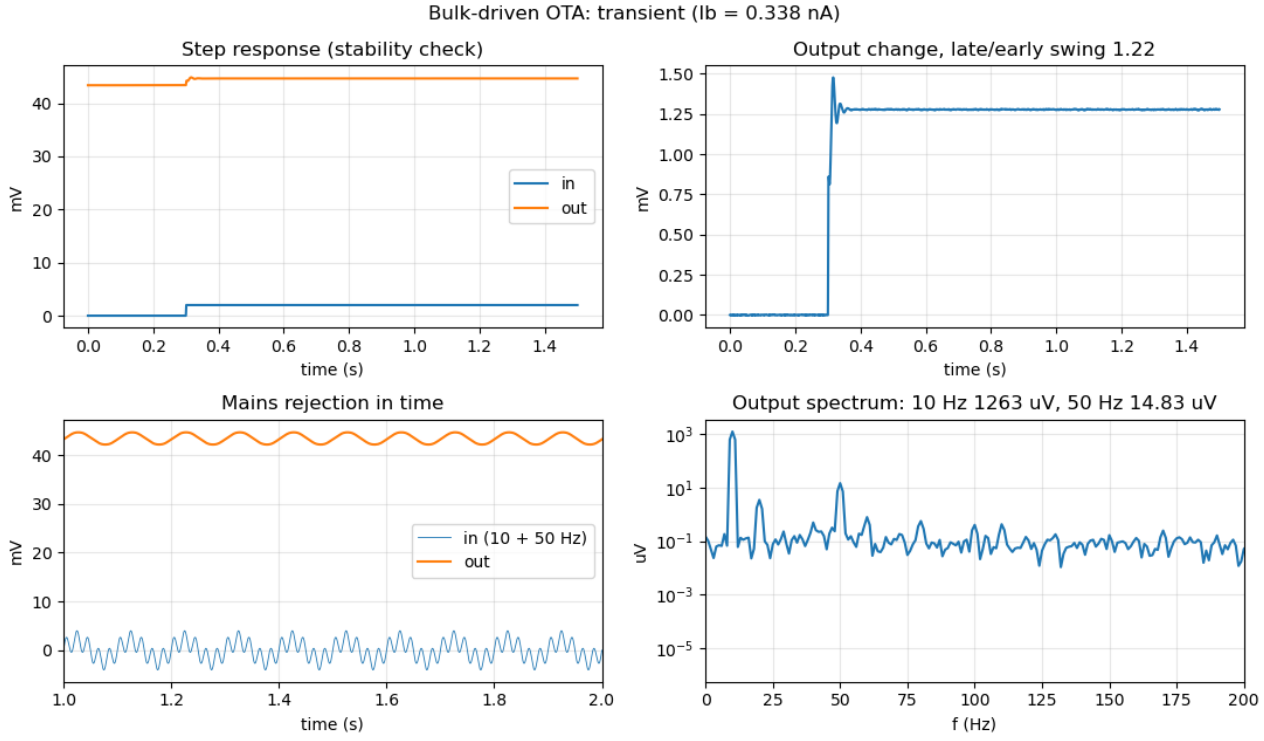


Figure 5. Step response and 10 + 50 Hz mains-rejection test.

7. Monte Carlo, corners and distortion

Mismatch Monte Carlo, 100 runs. Every run kept a notch of at least 20 dB, against 7 failures for the current-cancellation style. The cost is a wider frequency spread: f_0 = 50.25 ± 3.15 Hz for the 98 runs with normal passband gain, 57% within ±2 Hz and 92% within ±5 Hz; depth 37.9 ± 3.6 dB. The wide linear range keeps every OTA working even when offsets shift its operating point; the weak body control turns the same device mismatch into a larger g_m spread.

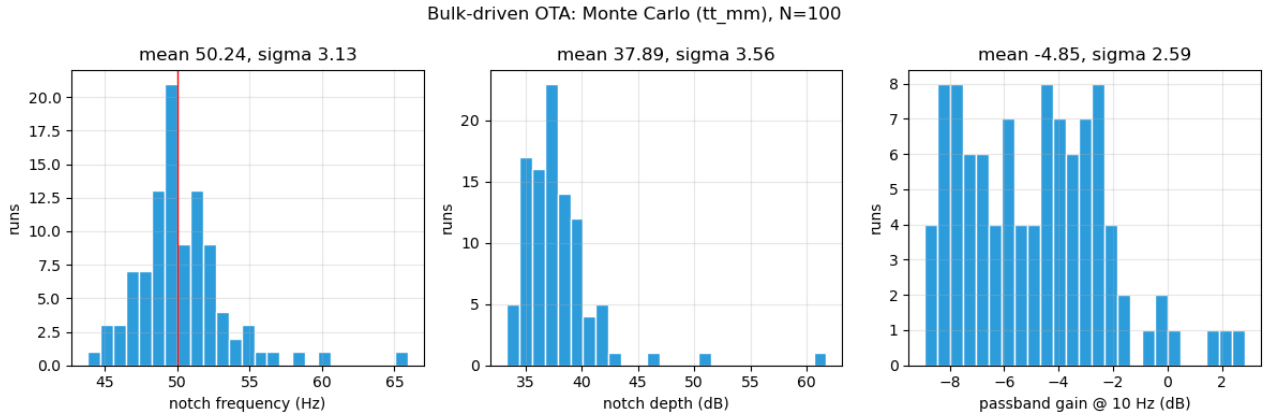


Figure 6. Monte Carlo histograms (mismatch, 100 runs).

Corner	T (°C)	f_0 at fixed Ib (Hz)	Depth (dB)	Gain 10 Hz (dB)	Ib trimmed (nA)	f_0 trimmed (Hz)
tt	27	49.97	38.2	−4.00	0.3385	49.99
ss	27	46.63	43.2	−6.37	0.3627	49.69
ff	27	53.91	34.4	−8.19	0.3137	50.36
sf	27	45.58	41.6	−8.76	0.3710	49.66
fs	27	55.13	36.8	−8.39	0.3067	50.37
tt	37	49.11	37.5	−6.72	0.3443	49.94
tt	0	52.28	40.5	−4.09	0.3235	50.22
tt	85	44.85	41.8	−8.24	0.3771	49.67

Bias trimming brings every corner back within 0.4 Hz of 50 Hz. As in the first style, the passband gain moves noticeably with corner, here between −4.0 and −8.8 dB.

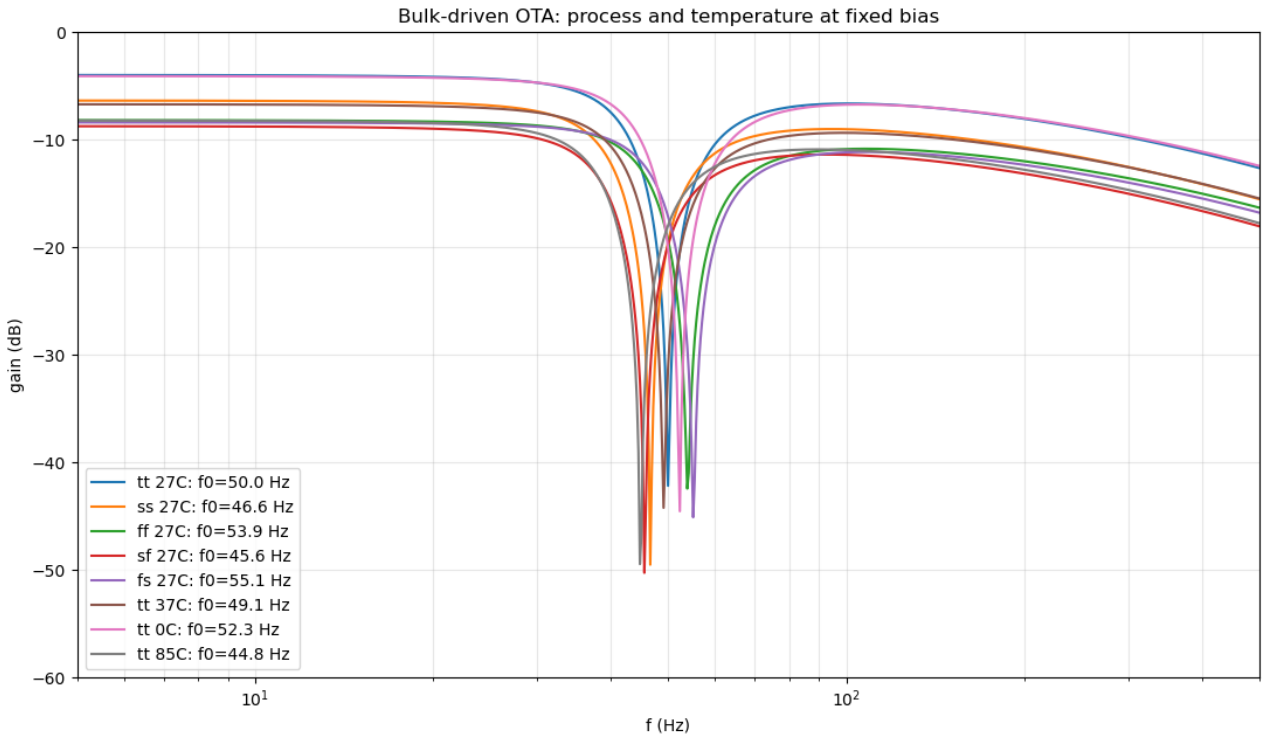


Figure 7. Response across process corners and temperature at fixed bias.

Input (mV peak)	Output fundamental (mV)	THD at output (%)	THD at filter output (%)	HD2 at output (%)	HD3 at output (%)
5	3.157	0.814	0.070	0.811	0.017
10	6.323	1.610	0.039	1.609	0.051
20	12.703	3.220	0.044	3.214	0.196

The filter itself is very linear (0.04–0.07% THD before the buffer), thanks to the wide linear range of the bulk-driven OTA. Almost all the distortion is second harmonic added by the buffer. This filter's output sits at +43 mV DC, which lands on the S-shaped region of the buffer's transfer curve near +0.1 V (visible in the buffer validation). The 5 mV target passes at 0.81%; 20 mV fails at 3.2%. Re-centring or linearising the buffer is the fix, and it is shared by all three styles.

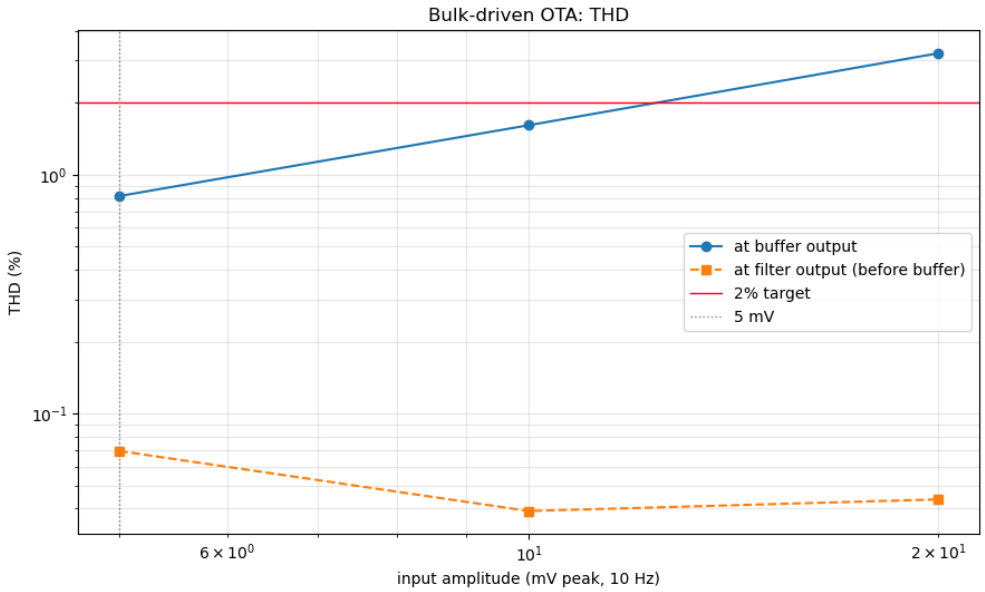


Figure 8. THD vs input amplitude: at the buffer output and at the filter output.

8. Power

Static power is P = 1.8 V × total current. Here the bias network has an extra branch: the vg replica carries Ib/2.

$I_{filter} = 5 \times I_{tail} + I_b + I_b/2 = 5 \times 0.3541 + 1.5 \times 0.3382 = 2.278$ nA

$P_{filter} = 1.8$ V × 2.278 nA = **4.10 nW**

P_{total} = 1.8 V × 874.5 nA (full testbench) = **1.574 µW**, of which the buffer is 1.565 µW

The bulk-driven filter uses about 1.8 times the current of the current-cancellation filter (4.10 vs 2.32 nW), because its body-controlled g_m per ampere is lower. Both are negligible next to the buffer.

9. Timing

Quantity	Method	Value
Integrator time constant	$\tau = C_{L2}/g_m$ = 4.084 pF / 1.379 nA/V	2.96 ms
Group delay	$-d\varphi/d\omega$ from the AC phase	2.52 ms at 1 Hz; 2.74 ms at 10 Hz; 3.59 ms at 20 Hz; 5.95 ms at 30 Hz
Step rise 10–90%	2 mV step	9.57 ms
Settling to ±2%	2 mV step	40.8 ms
Overshoot	2 mV step	15.6%
OTA maximum output current	All tail current to one side: $I_{max} = I_{tail}$	354 pA
Slew rate on C_{L2}	$SR = I_{max}/C_{L2}$	86.7 V/s
Slew-limited amplitude at 50 Hz	$SR/(2\pi \times 50$ Hz)	276 mV peak

Slew rate is three times higher than in the current-cancellation OTA, because there is no cancellation: the whole tail current is available at the output.

10. Layout

The bulk-driven OTA cannot use the interleaved array of the first style, because each input transistor's n-well is the input. The floorplan is therefore:

• **PMOS row, four separate n-wells** spaced at least 1.27 µm apart (the rule for wells at different voltages): [M7, Mt] at VDD | [M1] = inn | [M2] = inp | [M8] at VDD. Each well has its own n-tap strip; the inn and inp pins sit on the tap strips of their wells.

• **Bus band** on metal 1 below the PMOS row (vdd, cs, d1, d2, p7, out); gate buses above the row (vg, vb, p7).

• **NMOS row** below the band with gates facing up into it and sources down to a VSS substrate tap.

M1 and M2 sit side by side with identical surroundings, but there is no interleaving or common centroid, because each needs its own well. Splitting each into two units in two wells, connected in metal, would be the next step for matching.

Verification. The first pass had 10 DRC errors: a VDD connector sat 0.03 µm from the tap strip instead of touching it, which also left the first n-well floating and failed LVS. The second pass had 4 errors: the short p7 bus stopped 0.24 µm short of M7's drain contact. Both were fixed in the generator. Final result: **DRC 0 errors, LVS circuits match uniquely**. The layout is 90.2 × 27.4 µm, about 2,470 µm², roughly 40% smaller than the current-cancellation OTA.

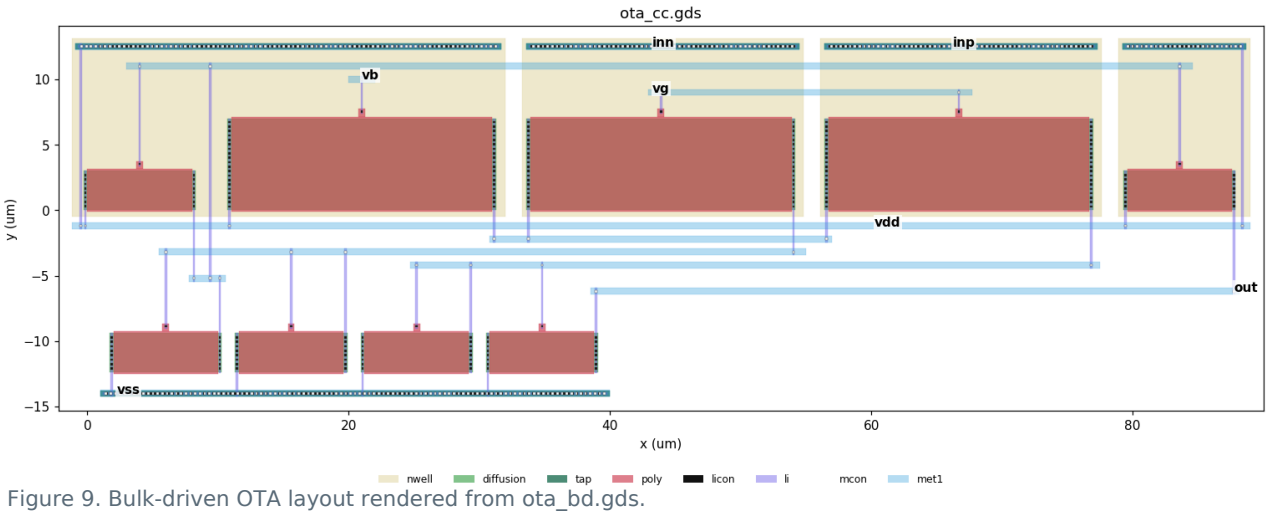


Figure 9. Bulk-driven OTA layout rendered from ota_bd.gds.

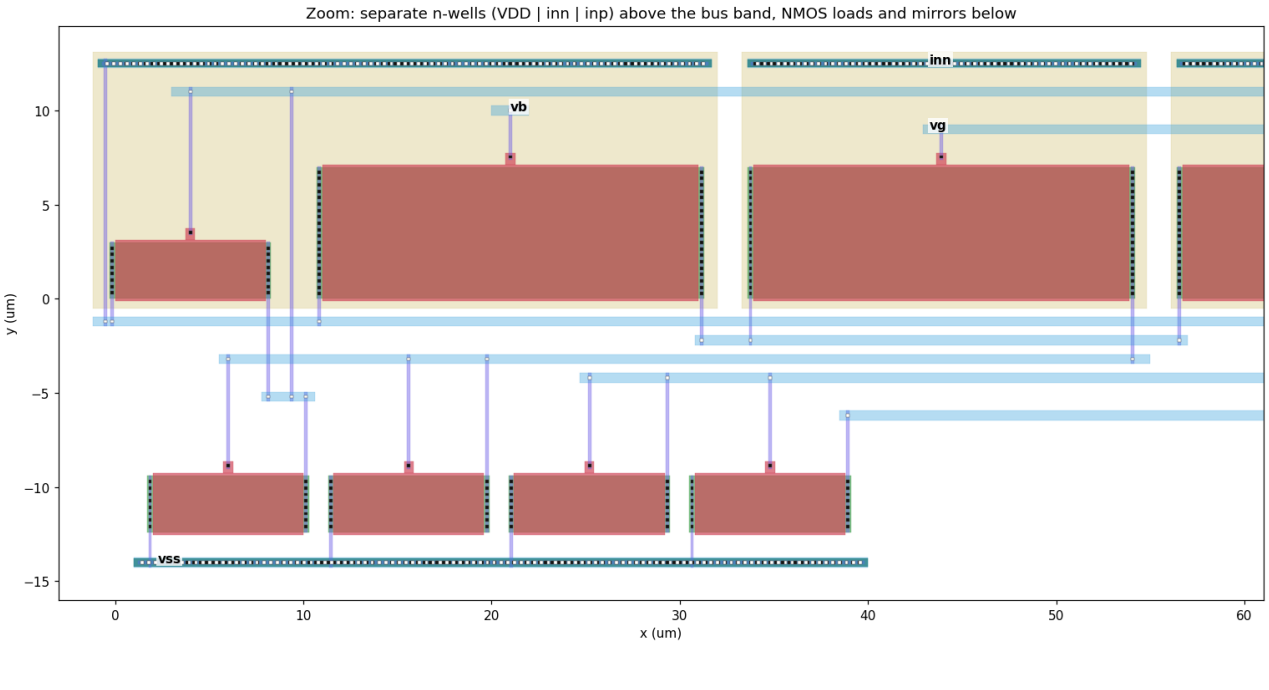


Figure 10. Zoom: the separate n-wells of the PMOS row, the bus band and the NMOS row.

11. Post-layout analysis

Extraction gives 211.7 fF over 52 capacitors per OTA. The largest signal-related terms are **36.4 fF from inp to VSS and 36.4 fF from inn to VSS**: the bottom capacitance of each input n-well to the substrate. This is the characteristic cost of bulk driving. That capacitance sits directly on the filter's internal nodes, through the inputs of the OTAs connected to them.

Quantity	Schematic	Extracted layout
Notch at the pre-layout bias	49.97 Hz	49.47 Hz (−0.5 Hz)
Bias for 50 Hz	0.3382 nA	0.3418 nA (+1.1%)
Bias for 60 Hz	0.4133 nA	0.4176 nA
Notch depth 50 / 60 Hz	38.2 / 38.1 dB	34.3 / 34.3 dB
Gain at 10 / 100 / 150 Hz	-4.00 / -6.62 / -7.17 dB	-4.00 / -6.85 / -7.47 dB
THD at 5 / 10 / 20 mV	0.81 / 1.61 / 3.22 %	0.80 / 1.60 / 3.22 %
Monte Carlo usable runs	98 / 100	100 / 100
Monte Carlo f_0	50.25 ± 3.15 Hz	50.08 ± 2.38 Hz
Monte Carlo depth	37.9 ± 3.6 dB	34.1 ± 1.6 dB
f_0 within 50 ± 5 Hz	92%	95%

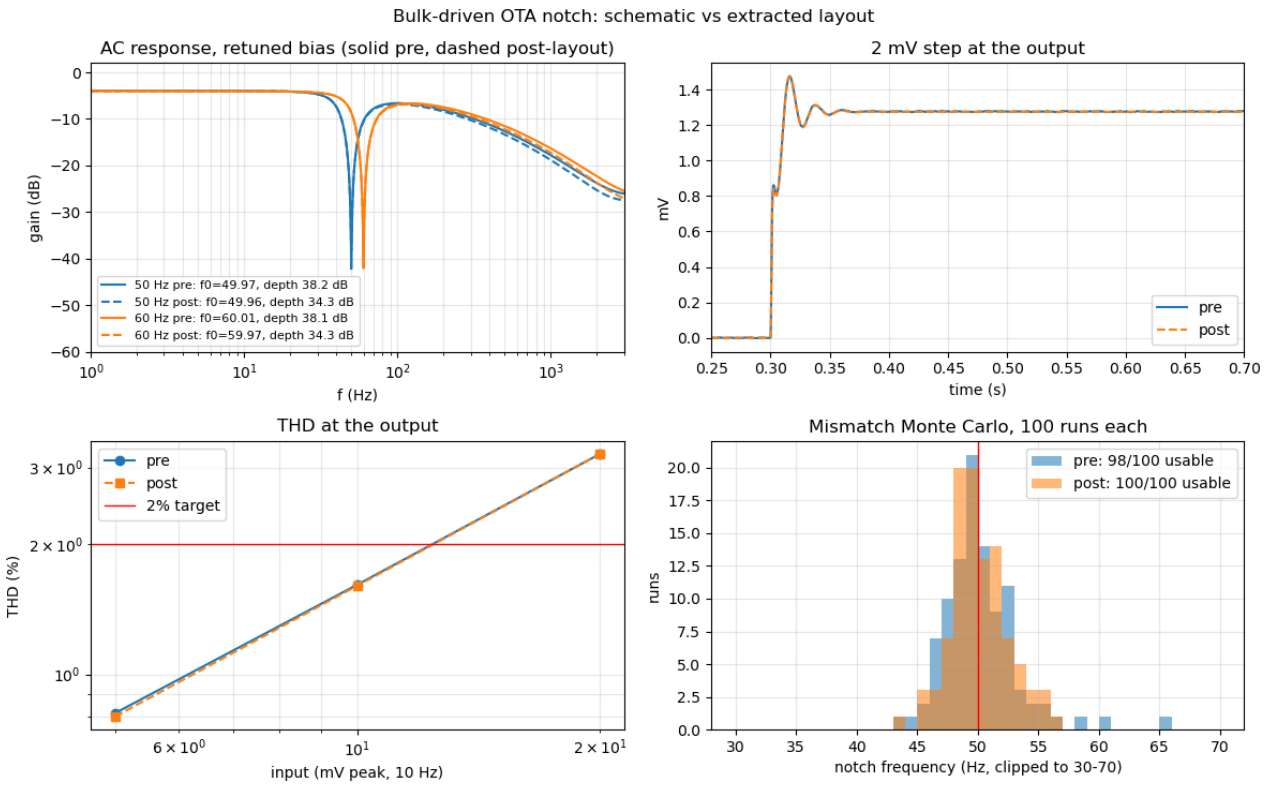


Figure 11. Schematic (solid) vs extracted layout (dashed): AC, step, THD and Monte Carlo.

Layout moves the notch by 0.5 Hz, which a 1.1% bias change corrects. Depth drops by about 4 dB, compared with about 1 dB for the current-cancellation OTA, which is consistent with the extra n-well capacitance on the input nodes. The two Monte Carlo runs are not paired sample by sample, so small differences in yield are not meaningful.

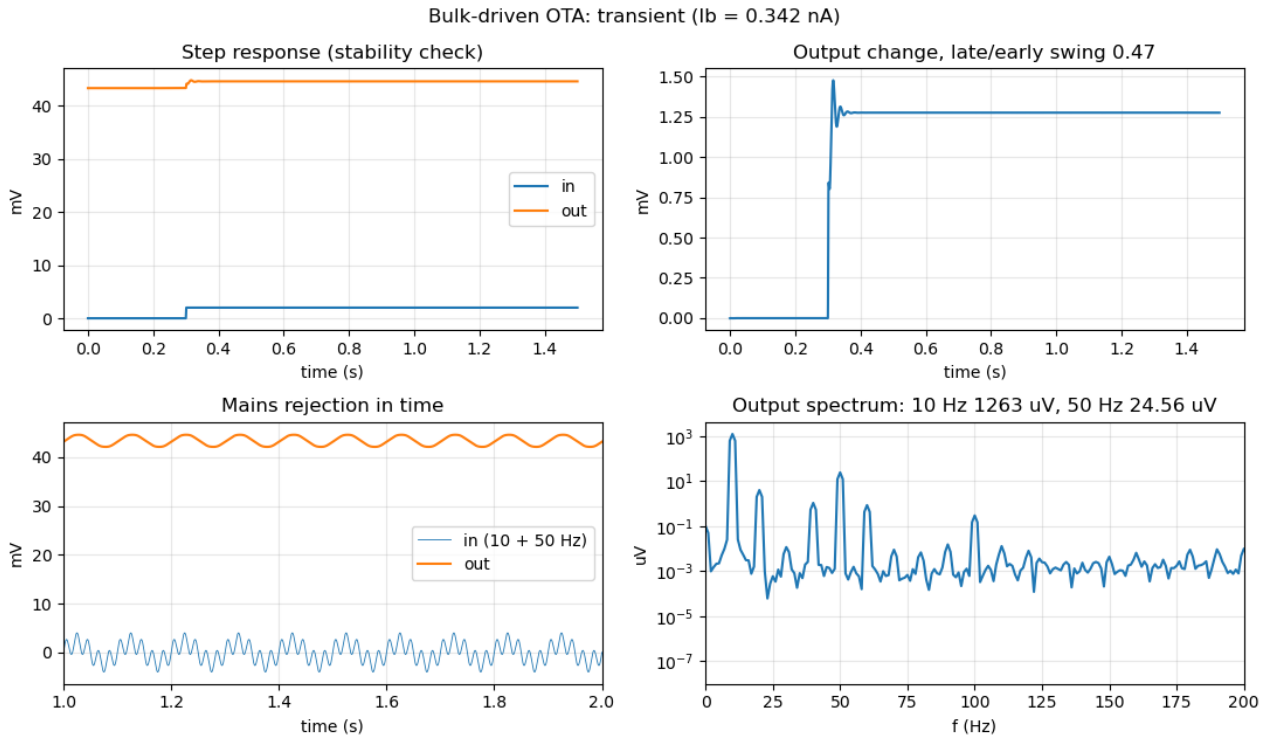


Figure 12. Post-layout transient.

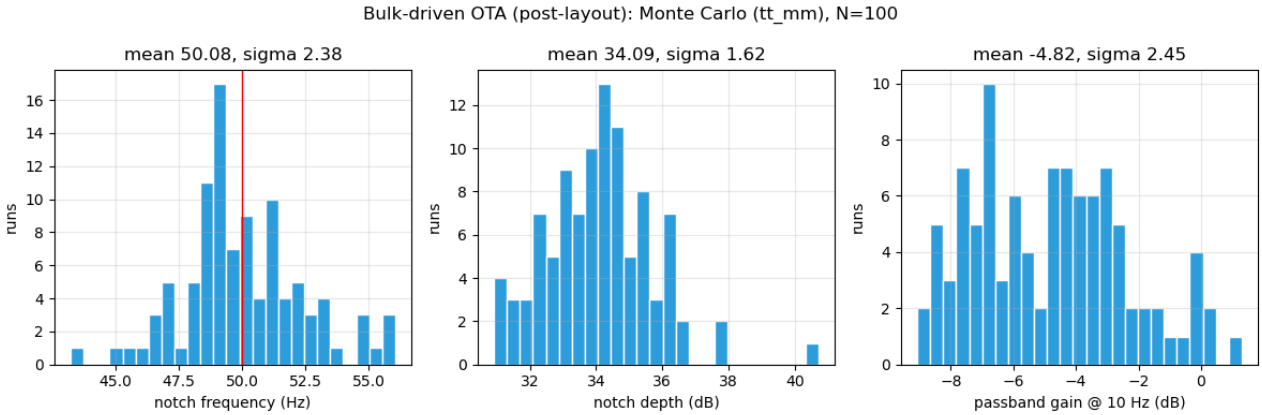


Figure 13. Post-layout Monte Carlo (100 runs, mismatch).

12. Limitations

• **Buffer distortion** limits THD above about 12 mV; the filter itself is far more linear.

• **f_0 spread** under mismatch is larger than in the first style (σ about 3 Hz), so a trim of the bias would be needed per chip.

• **Passband gain** varies with corner (−4.0 to −8.8 dB); the explanation in section 6 is likely, not proven.

• **Noise** is about twice that of the current-cancellation OTA (313 µV rms input-referred).

• **Process Monte Carlo** is covered by corners only (the PDK's mc section fails); the bias is an ideal current; only the OTA is laid out; the input pair is not common-centroid and has no dummies.